



CAPSTONE PROJECT 1

E-COMMERCE CUSTOMER INTELLIGENCE

CONVERSION ANALYTICS & PURCHASE PREDICTION

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PROBLEM STATEMENT

E-commerce businesses generate large volumes of data across customer interactions, website sessions, product engagement, cart activity, and transactions. These signals are often analyzed in isolation, making the complete customer journey to purchase difficult to understand. Businesses struggle to identify:

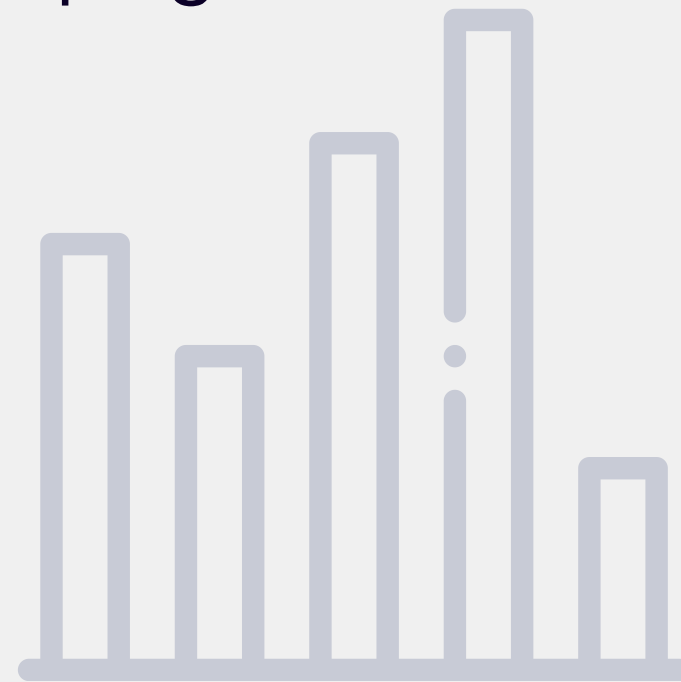
- where customers disengage
- which behaviors drive conversion
- which product, device, engagement, and promotional factors influence purchase

PROJECT OBJECTIVE

Build a unified analytical framework to understand customer behavior, conversion patterns, and predict purchase/conversion using only information available before the purchase outcome.

DATASET OVERVIEW

- **Dataset** | Kshashtra - ECommerce Store Marketing & Sales
- **Source** | Kaggle
- **Structure** | 9 tables
- **Total Records** | 599,960
- **Business Coverage** | Customers, Website Behavior, Products, Orders, Inventory, Purchase Orders & Marketing Campaigns



SQL | DATA AUDIT & INTEGRATION

9 Raw Tables → Audited → Cleaned → Integrated → 4 Processed Tables

- Created cleaned versions of the source tables by standardizing data types, dates, text fields, and monetary values.
- Joined and aggregated to create analytical datasets connecting customers, orders, products, inventory, supply, and marketing.
- Defined one website session as one observation and controlled data leakage by excluding post-purchase information.

THE FINAL 4 ANALYTICAL/ML TABLES CREATED

- **customer_order_product (30,000, 33)** = Customer + Order + Product/SKU view
- **sku_inventory_supply (11,495, 13)** = SKU + Inventory + Purchase/Supply view
- **marketing_website_summary (21,915, 24)** = Marketing + Website performance view
- **purchase_prediction_ml (26,611, 12)** = Session-level purchase prediction dataset

PYTHON | DATA PREPARATION & EXPLORATORY ANALYSIS

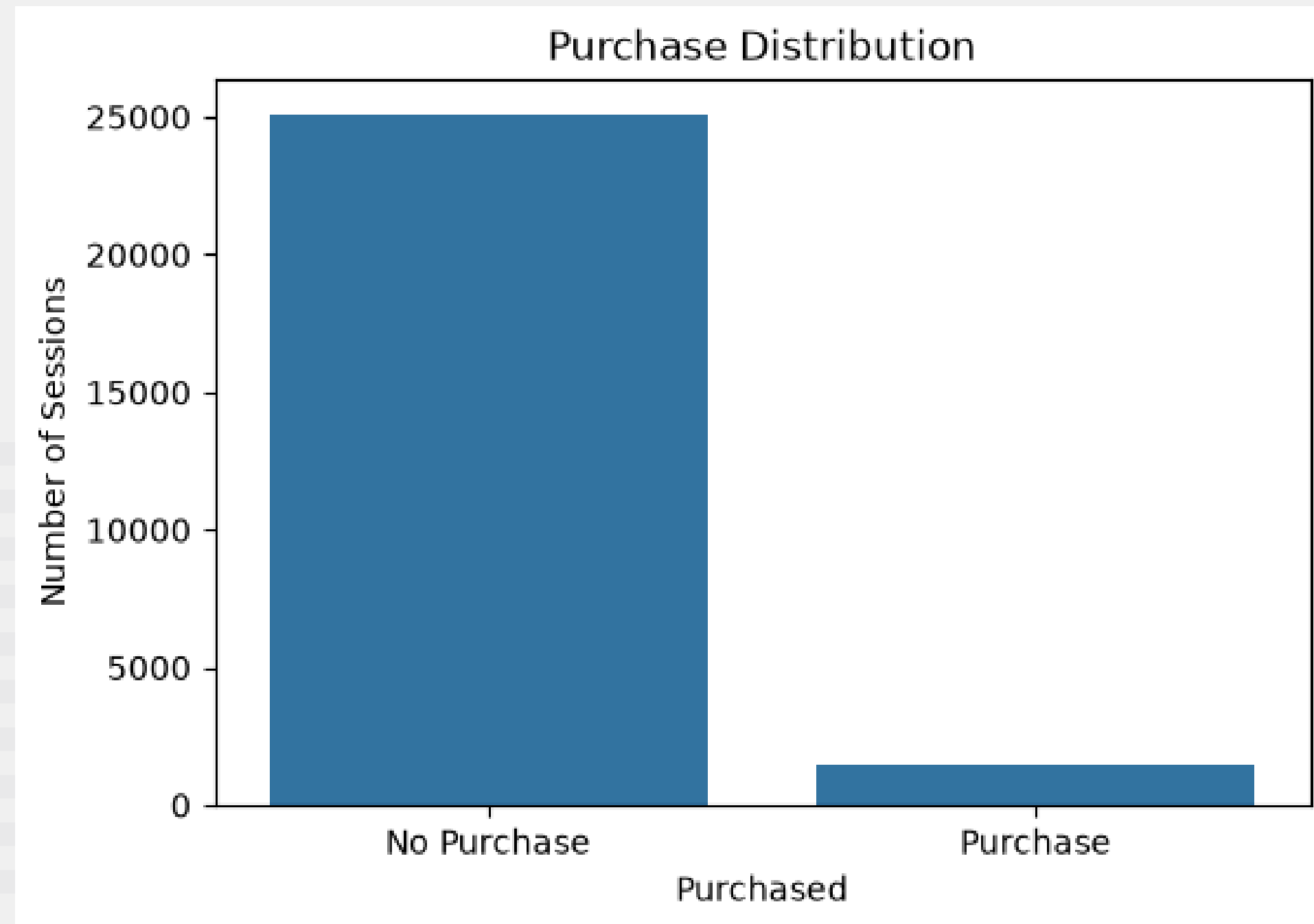
Load → Inspect → Clean → EDA → Feature Engineering → ML

- Exploratory Data Analysis (EDA) is performed to understand the structure, distribution, patterns, and relationships within the dataset.
- The main objective is to understand which session-level behaviors and contextual variables are associated with purchase conversion.

ML Dataset EDA | Understanding Purchase Behavior

- 26,611 Sessions
- 1,501 Purchases
- 5.64% Purchase Rate

DATA PREPARATION & EXPLORATORY ANALYSIS



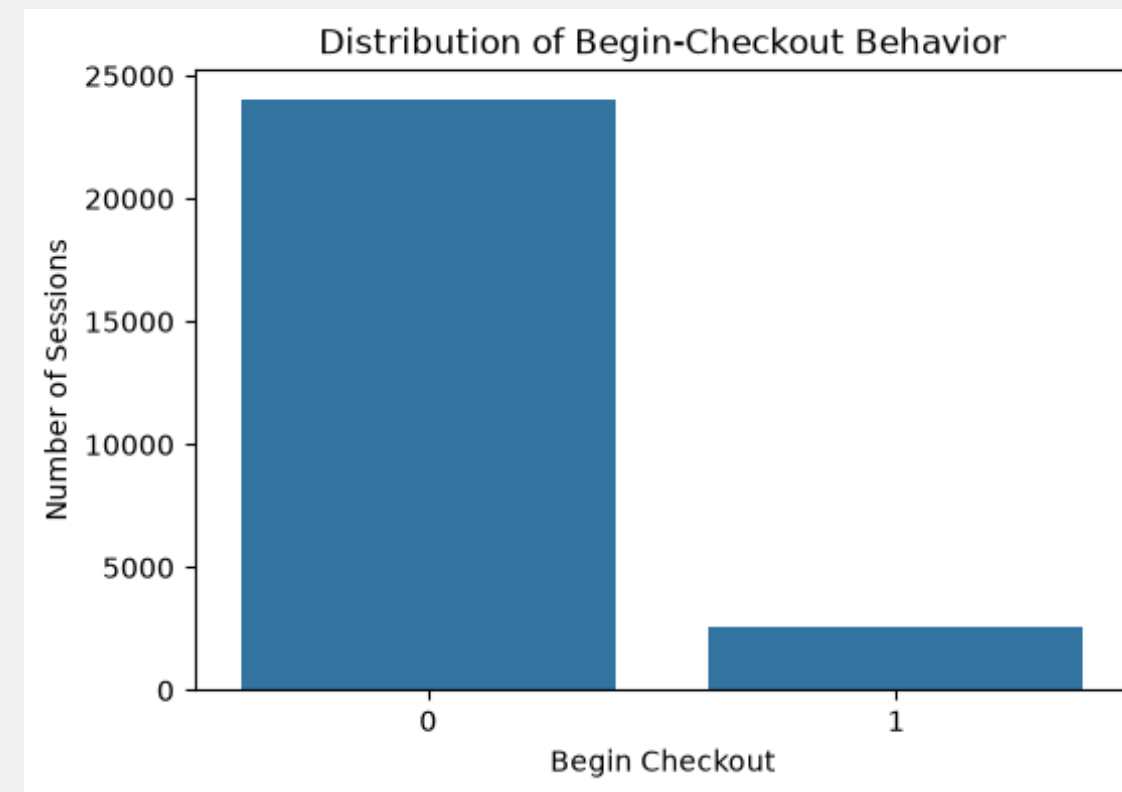
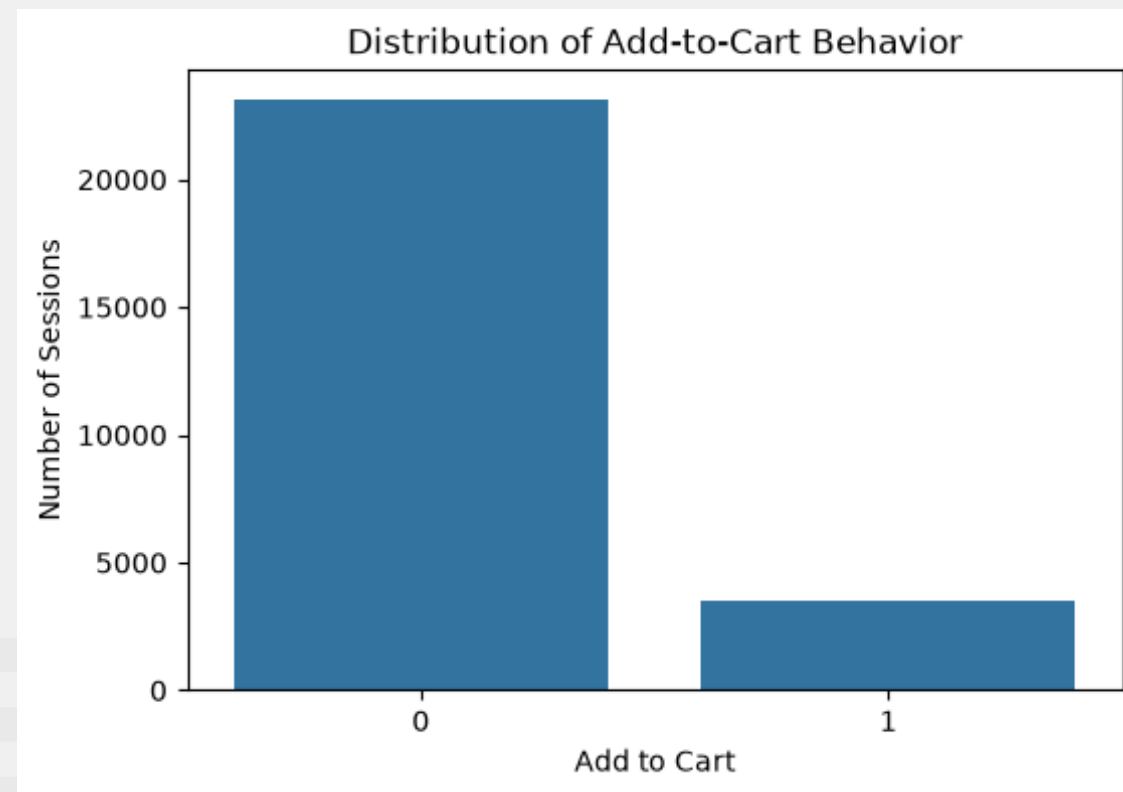
Purchase Target Distribution

- Purchased = 0 → 94.36%
- Purchased = 1 → 5.64%

Insight

- The target is highly imbalanced.

DATA PREPARATION & EXPLORATORY ANALYSIS



Insight

- 13.14% sessions reach cart
- 74.18% of carts reach checkout
- 57.86% of checkouts purchase
- The largest opportunity is earlier in the funnel - moving browsers into cart/checkout.

Conversion Funnel

26,611 Sessions



3,497 Add to Cart

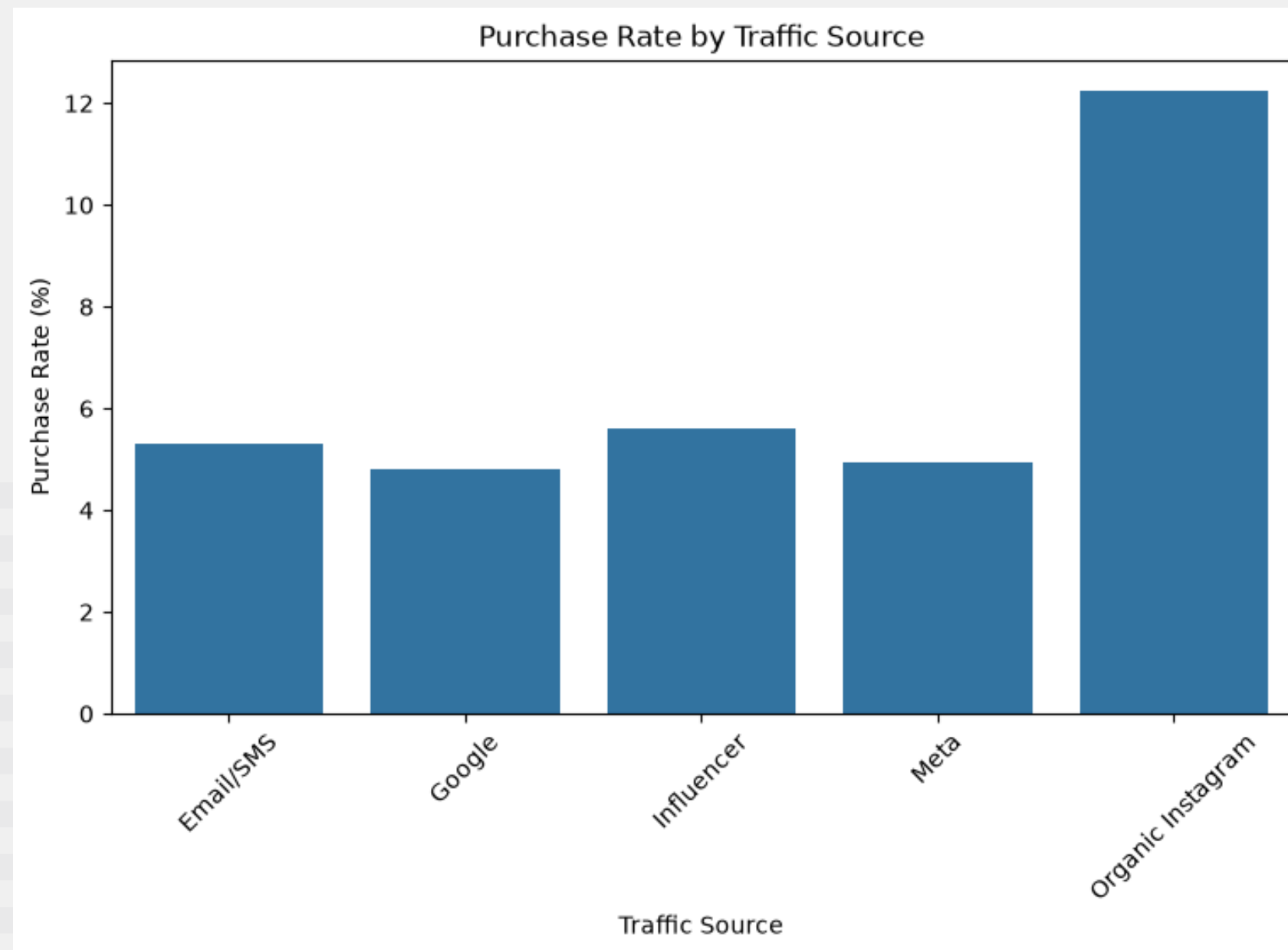


2,594 Begin Checkout



1,501 Purchases

DATA PREPARATION & EXPLORATORY ANALYSIS



Purchase Rate by Traffic Source

- Meta generates the largest volume of sessions and purchases.
- Organic Instagram has the highest overall purchase rate at 12.23%.

FEATURE ENGINEERING | REPRESENTING PURCHASE INTENT WITHOUT LEAKAGE

Feature	Business meaning
Product Views	Browsing intensity
Add to Cart	Product intent
Begin Checkout	Strongest observed purchase intent
Traffic Source	Acquisition context
Device + City	Session context

EXCLUDED FROM PREDICTORS

- session_id → identifier
- customer_id → identifier / unavailable for many visitors
- order_id → outcome-related
- revenue → post-purchase information
- post-purchase price/order variables

PREPROCESSING

- Handle missing values
- One-Hot Encoding categorical variables
- Standard Scaling numerical variables
- Train/test split

MODEL BUILDING & 5-FOLD STRATIFIED CROSS-VALIDATION

Model	Accuracy Mean	Accuracy Std	Precision Mean	Recall Mean	F1 Mean	ROC-AUC Mean
Random Forest	95.74%	0.17%	61.73%	64.36%	62.99%	98.05%
Balanced Random Forest	95.99%	0.24%	59.76%	88.76%	71.40%	98.01%
Balanced Logistic Regression	95.87%	0.26%	57.78%	100.00%	73.23%	98.00%
Logistic Regression	95.60%	0.27%	57.49%	85.93%	68.81%	98.00%
Decision Tree	95.56%	0.11%	61.54%	56.62%	58.97%	82.36%
Balanced Decision Tree	95.63%	0.18%	60.35%	65.78%	62.92%	81.60%

MODEL SELECTED: BALANCED LOGISTIC REGRESSION

Why?

- 100% Recall - no observed purchasers missed
- 73.23% F1 - highest among the six models
- 98.00% ROC-AUC - strong class discrimination
- Suitable for the imbalanced purchase target

Cross-Validation:

- Accuracy: 95.87% $\pm$ 0.26%
- Precision: 57.78% $\pm$ 1.54%
- Recall: 100.00% $\pm$ 0.00%
- F1 Score: 73.23% $\pm$ 1.24%
- ROC-AUC: 98.00% $\pm$ 0.16%

BUSINESS INTELLIGENCE & DASHBOARDING

- The three analytical tables: Customer Order Product, SKU Inventory Supply, and Marketing Website were further analyzed to derive key business insights and performance trends.
- Power BI was used to develop interactive dashboards covering customer behavior, sales and product performance, inventory and supply, and marketing and website performance.
- Excel was also used for supporting analysis, validation, and business-level data exploration.
- Together, these tools helped transform the processed data into visual, actionable insights for decision-making.

OVERALL FINDINGS & BUSINESS RECOMMENDATIONS

Finding	Business Recommendation	Expected Impact
Repeat customers have higher AOV ~₹2.8K vs ~₹2.4K for first-time customers	Build post-purchase journeys, personalized recommendations, loyalty offers, and reactivation campaigns	Increase customer retention & lifetime value
Checkout → Purchase conversion is 57.86%	Reduce cart and checkout friction and prioritize high-intent sessions	Improve conversion rate & revenue
Shirts are the strongest category ~₹26M revenue	Protect availability of winning SKUs and test bundles/cross-selling	Protect revenue & increase basket value
Meta provides scale; Organic Instagram shows stronger conversion efficiency	Optimize marketing using traffic volume + conversion efficiency + customer value , not traffic alone	Improve marketing productivity
Inventory coverage appears very high (~484–596 days for displayed top SKUs)	Use sales velocity + supplier lead time for demand-led replenishment; validate snapshot freshness	Reduce overstock & stockout risk
ML achieved 100% recall on the test set	Use purchase probability to prioritize high-intent sessions for retargeting/personalization	Improve targeting efficiency & lead prioritization

CONCLUSION & FUTURE SCOPE

Conclusion

- Built an end-to-end E-commerce Customer Intelligence framework.
- Combined SQL, Python, Power BI & Excel for analysis and decision support.
- Identified key customer, funnel, product, marketing & inventory insights.
- Selected Balanced Logistic Regression with 100% Recall and 73.23% F1.

Future Scope

- Real-time purchase prediction & personalization
- Customer segmentation, CLV & churn prediction
- Improved marketing attribution & ROAS
- Advanced demand forecasting & inventory optimization
- Continuous model and KPI monitoring

THANK YOU

E-Commerce Customer Intelligence Conversion Analytics &
Purchase Prediction